INCH-POUND
MIL-DTL-24643/37F
1 October 2009
SUPERSEDING
MIL-DTL-24643/37E
22 August 2002

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, -20 °C TO +90 °C, TYPE LS3U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

Construction, Non-Watertight

First	-	Copper conductor, coated or uncoated, 18 AWG size, Class B stranding (7 by 0.0152 inches).
Second	-	Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third	-	Thermoset insulation. Nominal wall thickness of 0.023 inch; wall thickness may vary from nominal as necessary, in order for the completed cable to meet the specified electrical requirements. Colored insulation, one white, one black, and one red conductor for each triad.
Fourth	-	Three conductors (one black, one white, and one red) cabled together with a nominal lay of 3.5 inches to form a triad.
Fifth	-	Identification braid on each triad. Standard identification code applied by Method 4.
Sixth	-	The specified number of triads (see table I) shall be cabled together in accordance with 3.4.4 of MIL-DTL-24643. Cabling sequence shall be consecutive starting with no. 1 from the center outward. Fillers may be used as necessary to form a firm well-rounded assembly.
Seventh	-	An optional binder.
Eighth	-	Cross-linked polyolefin jacket (see table I for thickness)

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TABLE I. Details.

Military part no. M24643/37	Type and size	Number of triads	Cable jacket thickness min. avg. (inch)	Overall diameter	
				min. (inch)	max. (inch)
-01UN	LS3U-3	3	0.040	0.580	0.620
-02UN	LS3U-7	7	0.040	0.760	0.810
-03UN	LS3U-12	12	0.050	1.030	1.090

REQUIREMENTS:

Qualification required.

INSPECTION:

Basic Electricals:

Conductor resistance (ohms/1000 feet at 25 °C, max.)	7.06
Voltage withstand (volts, root mean square, min.)	
Conductor to conductor	2000
Insulation resistance (megohms/1000 feet, min.)	
Conductor to conductor	500
Conductor continuity	No failure

Group A:

Visual and dimensional	No failure
Capacitance	
Mutual capacitance, 1 kilohertz (KHz) (pF/ft, max.)	30
Capacitance unbalanced (percent max.)	8

Group B:

Thermoset proof test (percent, max.)	
Insulation	50
Jacket (when tested at 200 °C)	50
Physicals (unaged)	
Insulation (extruded)	
Tensile strength (lb/in^2, min.)	700
Elongation (percent, min.)	150
Jacket (cable)	
Tensile strength (lb/in^2, min.)	1300
Elongation (percent, min.)	160

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Tear (lb/in thickness, min.)	35
Group C:	
Physicals (aged) air oven	
Insulation (extruded)	
Tensile strength (percent of unaged, min.)	75
Elongation (percent of unaged, min.)	75
Jacket (cable)	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60
Permanence of printing (jacket) (cycles, min.)	125
Heat distortion (percent of unaged, max.)	30
Shrinkage	No failure
Group D:	
Flame propagation (cable)	No failure
Qualification Inspection:	
Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:	
Aging and compatibility (cable) (125±5 °C)	No failure
Abrasion resistance (jacket) (scrapes, min.)	75
Acid gas equivalent (percent, max.)	
Jacket	2
Fillers	2
Insulation	18
Halogen content (percent, max.)	
Jacket	0.2
Fillers	0.2
Insulation	0.2
Immersion (jacket)	
Tensile strength (percent of unaged, min.)	50
Elongation (percent of unaged, min.)	50
Smoke index (max.)	
Jacket	25
Fillers	45
Insulation	45

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Toxicity index (max.)	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - Type A (hardness, min.)	80
Weathering (jacket)	No failure
Electrical moisture absorption	No Failure

UNIT ORDERING LENGTHS:

Type and size	Feet (nominal)
LS3U all sizes	1000

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army – MI
Navy – SH

Preparing Activity:
Navy – SH
(Project 6145-2008-041)

Review Activities:
Army – AR, AV, CR
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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